

An Access Control Mechanism in Smart Identifier Network

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Abstract—In recent years, with the development of mobile Internet, cloud computing, and other technologies, the traditional Internet architecture has been challenging to meet the needs of future network development. As a revolutionary network architecture, SINET provides security and mobility for the network through the identification mapping separation technology. At the same time, the development of programmable data plane technology provides conditions for the large-scale deployment of SINET. In this paper, we proposed an Attribute-based access control(ABAC) mechanism based on multidimensional attributes of users and services in the PDP-based SINET. We designed the workflow of property registration and access requests of the network. We built a network prototype system and verified the function of the access control mechanism. The results show that the access control mechanism proposed in this paper can meet the network's fine-grained security management and control requirements and ensure the secure access of users and services in the network.

Keywords—SINET, ABAC, programmable data plane, network security

I. INTRODUCTION

In the traditional Internet architecture, IP addresses have the characteristics of dual binding of identity and location, which leads to a series of problems in the existing network in terms of data transmission security, mobility, routing scalability, etc., making network user management challenging to achieve. Since the root cause of these problems is the traditional architecture design of the Internet, the use of protective measures such as system patches still cannot fundamentally solve the above problems. SINET uses the Access Identifier (AID) to represent the identity information of the terminal, the Routing Identifier (RID) to represent the location information of the terminal, and the decoupling of the terminal location and identity through the separation mapping resolution mechanism, which effectively solves the shortcomings of the existing network and provides a feasible alternative to the problems of poor support and security of traditional network mobility. [1]It has positive significance for future network development and is an influential next-generation Internet representative architecture.

However, because the current network environment has long relied on the traditional TCP/IP system, it is not easy to

directly realize the large-scale deployment of SINET. At this time, the development of Software-Defined Networking (SDN)[2] and programmable data planes represented by P4 has brought a new dawn to the deployment of integrated identity networks[3]. Unlike traditional data planes, which only support fixed protocols, programmable data planes break the hardware limitations of traditional data planes. It allows users to customize the logical processing and forwarding process of data packets by programmable switches through programming, realizing the protocol independence of data forwarding.

The proposal of the programmable platform provides conditions for the promotion of SINET. Network security control is one of the key research directions of SINET. There is little research on network security control of SINET based on the programmable platform. A security control mechanism is needed to solve the network information security problem of SINET under the programmable platform. At present, access control technology is one of the key strategies to achieve network security control. Since the seventies of the last century, access control technology has been continuously developed, typical of which are Discretionary Access Control (DAC) which allows subjects to grant permissions to other subjects[4], mandatory access control (MAC) by authoritative authorities to assign subject and object security levels [5] and Use the mapping relationship of user-role and role-permission to realize role-based access control(RBAC) that decouples users from permissions[6]. There are also access control models for various fields.[7-10]However, with the rapid development of cloud computing, big data, artificial intelligence, and other technologies, the number of network users and the types of network services continue to increase, and the multi-dimensional attributes of users and services are gradually refined. The static single traditional access control mechanism is challenging to cope with the new requirements brought by these changes, and a more fine-grained, flexible, and dynamic access control mechanism is required to achieve refined management of the network. Attribute-Based Access Control (ABAC) provides the flexibility to use the attributes of requesting users and service providers to determine their access rights by introducing subject and object attributes as elements for making access decisions. When a principal's attributes change, its permissions also change, making network control more flexible.[11]

Based on the research of SINET under the programmable platform, this paper designs an access control mechanism based on multi-dimensional attributes, aiming to realize refined network management. According to the overall architecture of the integrated identity network, a set of an access control system is designed, including a terminal client, a request processing module, a terminal management module, a policy matching module, and a policy management module, through the interaction between these five modules, refined management of network users and services is realized, and the policy management module is used to improve the scalability of the system. The workflow of the SINET access control mechanism is designed, the integrated identity network prototype system is built, and the functional verification of the designed access control system is carried out.

II. NETWORK ARCHITECTURE

A. Overall Design

The SINET under the programmable platform adopts the architecture of control and data separation, and the overall architecture of the network is divided into two parts, the control plane and the data plane, as shown in Figure 1. Among them, the control plane uses the controller to implement functions such as topology management, access control, and delivery flow tables. The data plane mainly divides the network into two parts: the access network and the core network. In the access network, AID is used to represent the identity information of the terminal, and RID is used in the core network to represent the location information of the terminal. It is necessary to build an $\langle \text{AID}, \text{RID} \rangle$ mapping system between the access network and the core network to realize the communication between different access networks.

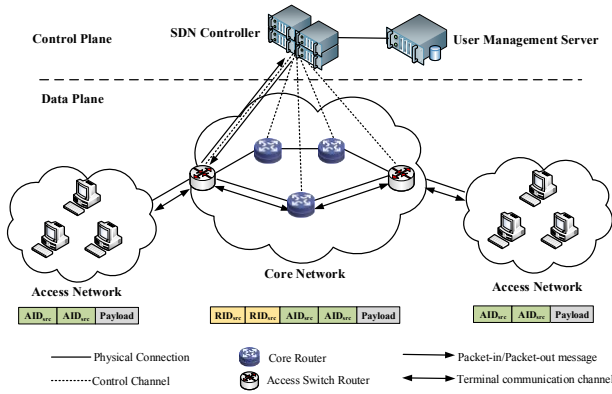


Fig. 1. The architecture overview.

In the network architecture, each terminal needs to register when it first enters the network. The system generates an exclusive AID for it according to its relevant information. The terminals use AID for resource sharing or network communication, and AID is bound to the device, which will not change with the movement of the terminal location.

At the same time, the controller will also generate the corresponding RID information for the terminal according to its location on the network. When a packet with AID information is transmitted to the access switching router, the

$\langle \text{AID}, \text{RID} \rangle$ mapping system maps the AID to RID for transmission in the core network.

B. Access Control System Design

Based on the overall architecture of the identity network under the programmable platform, combined with the attribute-based access control model, this paper designs an access control system framework based on multi-dimensional attributes of users and services, as shown in Figure 2, the system starts from the control plane and the data plane, and the data plane includes two modules: user/server client and request processing module; The control plane includes three modules: user management module, policy generation module, and policy management module, and its specific design is described below.

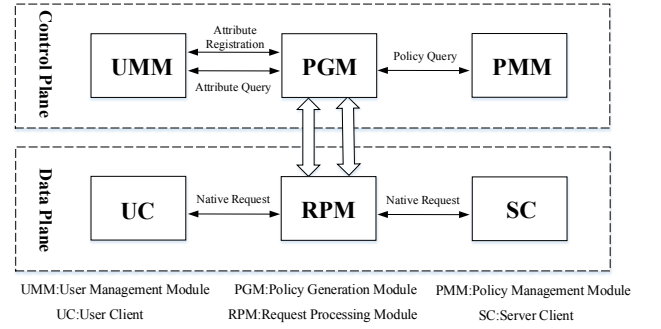


Fig. 2. The model of access control system.

III. MODULAR DESIGN OF ACCESS CONTROL SYSTEM

A. User/Server client

As the subject in the access system, the user/server client is mainly responsible for implementing the attribute registration function and the identity authentication login function, and the module runs in the terminal device. When a user logs in for the first time, they need to register their identity through the software first. In addition, when the user initiates an attribute registration request and logs in, the module will also send the attribute information related to the login device contained in the request to the terminal management module to obtain information such as the user's geographic location.

B. Request Processing Module

As a decision point in the access system, the terminal request processing module is mainly responsible for the user/service attribute registration request and access request processing, which is one of the key modules to implement the SINET access control mechanism $\langle \text{AID}, \text{RID} \rangle$ mapping system. In this access control system, this module serves as the policy execution point to complete the execution of user access policies and deny access to unauthorized users. At the same time, the module also needs to collect the user's access behavior information and provide data support for the subsequent judgment of the user's interest area and trust level. This module is deployed in the data plane's access switching router and is divided into the following three functions..

The data parsing function parses the integrated identification network packets transmitted from the terminal, obtains the packet type, and performs corresponding

forwarding operations. In the access control system, the main types of packets identified in the network are attribute registration requests and access requests. For attribute registration messages, extract the AID_{src} in the message and the related multidimensional attributes and then send the registration message to the terminal management module. Access rights are obtained for access requests, and operations are performed based on information such as the AID_{src} , the AID_{dst} , and the port number, to control users and service attributes.

The policy query and execution function is used to query the policy of terminal access request packets and execute access policies. Suppose the access switch router cannot determine the access rights of the terminal that initiated the access request. In that case, it sends an access policy query packet to the policy matching module, determines the access rights of the terminal based on the response flow table items issued by the policy matching module, and executes the corresponding access policy.

The mapping conversion function completes the mapping conversion between $\langle AID, RID \rangle$. When the request packet is allowed to access the destination network, and the destination terminal is not on the same access network as the requesting terminal, ASR needs to map the AID to RID to facilitate the transmission of the packet in the core network. After the policy matching module completes the judgment of the access request, it will query the access request allowed to access the destination network and send the $\langle AID, RID \rangle$ mapping relationship to the access switching router.

C. User Management Module

The user management module is mainly responsible for storing user attributes, service attributes, access policies, mapping relationships, etc., which are used to manage user and service terminals.

For the attribute registration request sent by the request processing module, the module extracts the user's attribute information, location information, and login device information from the packet, obtains the RID based on the location information, generates an $\langle AID, RID \rangle$ mapping relationship for it, and stores the above information in the MYSQL relational database.

For mapping query requests, the module searches from the database based on AID information, obtains its corresponding RID information, and returns it. Suppose the AID information corresponding to the AID does not exist in the database. In that case, the terminal is considered connected to the network, and the query failure information is returned.

In addition, the module will analyze users and manage user behavior information by extracting the five-tuple information in user access behavior. It includes the service type with the most recent visits, resource security level, and user trust level. When the policy matching module queries the matching module, this information is sent to the matching module together with user attribute information and device information.

D. Policy Generation Module

The policy generation module is mainly responsible for matching the multi-dimensional attributes of users and services with the access policy in the system and sending the matching results of the access policy to the access switch router as a flow table.

After the module receives the access policy query request sent by the request processing module, it will query the relevant attributes of the terminal management module according to the user and service information extracted from the request, sort out the obtained user attributes and service attributes, query the access control policy in the current system with the policy management module, and determine the access request according to the policy. For requests that allow access, query the terminal management module for the $\langle AID, RID \rangle$ mapping relationship and send the mapping relationship to the access switching router along with the allowed access flow table. The access-denied flow table is directly issued for requests that deny access to prevent the request from entering the core network transmission. When the policy in the policy management module changes, the module needs to re-judge all users who have logged in to the user management module to ensure the timeliness of the access policy.

E. Policy Management Module

The policy management module mainly manages policies in the access control system and adds or deletes access policies. Access policies are stored as database entries in the SINET access control system designed in this article. When subsequent access policies need to be changed, this module can be used to modify the policies in the database to improve the system's scalability. This module combines the attributes of the terminal management module. It formulates more fine-grained access policies based on user trust, resource security, and other information to achieve advanced user management.

F. Access Control Workflow

After completing the authentication login, the user terminal initiates an access request to the service terminal. After receiving the access request packet, the access switch router extracts the source AID, destination AID, source port, destination port, and transmission protocol and matches the access control flow table. If the matching result is to allow access, the $\langle AID, RID \rangle$ mapping is performed directly, and the packets are forwarded to the core network router. If the matching result is that the access is denied, an access-denied packet is returned. If the match cannot be successful, the user terminal is considered to be accessed for the first time, and the access permission information of the terminal is not available in the access switch router. The access policy query needs to be queried, and the packet is encapsulated as a packet-in packet and sent to the controller. After receiving the packet, the policy matching module in the controller will query the user attributes, service attributes, and user behavior information from the terminal management module based on its AID information and judge the access request based on the existing access policies in the system. If the user is allowed to access the target service, the mapping relationship of $\langle AID, RID \rangle$ is queried to the terminal management module. Then the response packet, the allowed access flow table, and the mapping flow table are

delivered to the access switch router. When a subsequent user initiates an access request, Access Switch Router performs $\langle \text{AID}, \text{RID} \rangle$ mapping conversion on the user's packets and forwards the converted packets to the core network router for transmission. If the user is denied access to the target service, the response packet and the access-denied flow table are directly delivered to the access switching router.

IV. SIMULATION EXPERIMENT AND ANALYSIS

A. Experimental Environment

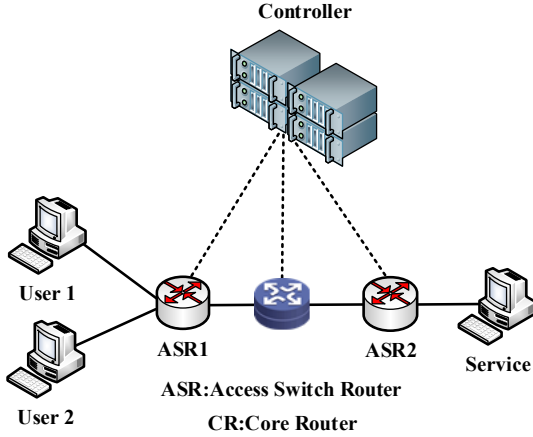


Fig. 3. The topology of the prototype system.

Figure 3 shows the topology of the identification network architecture based on the programmable platform. The control plane uses ONOS as the controller platform and runs on the Intel® Xeon® Silver 4212 CPU @ 2.20GHz server. Deploy applications to manage network topology and deliver flow tables. The data plane utilizes a Bmv2 software switch to parse and forward packets. The software switch is deployed on the Intel® Xeon® CPU X3440 @2.53GHz server. The user terminal uses the scapy module of the Python program to simulate the construction of identifying network packets and sending them. Detailed equipment parameters are summarized in Table I.

TABLE I. THE SUMMARY OF DEVICE PARAMETER

Device	Parameter	Count
User/Service	Intel Xeon E3-1230V2@ 3.30GHz 4.0 GB RAM	3
Controller	Intel Xeon Silver 4214 @ 2.2GHZ 64.0GB RAM	1
ASR/CR	Intel Xeon CPU X3440 @ 2.53GHz 16.0GB RAM	3

B. Simulation Result

To test the policy query and policy match, User 1 and User 2 initiate an access request to Service 1. After the policy matching module receives the access requests of two users, it obtains the attribute information of user 1 and user 2 and the attribute information of service 1 through the query. Among them, service 1 at the time of registration, the service industry category is "national defense industry", user 1 at the time of

registration industry is "military agency", and user 2 at the time of registration industry is "education industry". After receiving the access requests of the two users, the system matches the industries of the two users and the industry category of the service according to the access policy, determines that user 1 is allowed to access the target service and user 2 is not authorized to access the target service, and issues flow tables for different users to the programmable switch. Figure 4 shows a partial flow table in a programmable switch, from which it can be seen that user 1 has access permission and user 2 has access denied access. Through the above tests, the system can achieve fine-grained access control based on user attributes.

```

id=c9000084083cf4, state=ADDED, bytes=0, packets=0, duration=49, live
Type=UNKNOWN, priority=100, tableId=c_ingress.enable_srcrid, appId=SiNET,
selector=[hdr.aid.aid_s=0x20010001000000000000000000000000], treatment=D
efaultTrafficTreatment[immediate=[c_ingress.add_srcrid(rids=0x7327f54d)],
deferred=[], transition=None, meter=[], cleared=false, StatTrigger=null,
metadata=null]
id=c900009141706d, state=ADDED, bytes=0, packets=0, duration=45, live
Type=UNKNOWN, priority=100, tableId=c_ingress.enable_srcrid, appId=SiNET,
selector=[hdr.aid.aid_s=0x20010001000000000000000000000000], treatment=D
efaultTrafficTreatment[immediate=[c_ingress.add_srcrid(rids=0x7327f54d)],
deferred=[], transition=None, meter=[], cleared=false, StatTrigger=null,
metadata=null]
id=ca00006e78a322, state=ADDED, bytes=0, packets=0, duration=23, live
Type=UNKNOWN, priority=100, tableId=c_ingress.get_acl, appId=SiNET, se
lector=[hdr.aid.aid_s=0x20010001000000000000000000000000, hdr.aid.aid_d=0
x20010002000000000000000000000000], treatment=DefaultTrafficTreatment[im
ediate=[c_ingress.deny()], deferred=[], transition=None, meter=[], clear
ed=false, StatTrigger=null, metadata=null]
id=ca00009af3f858, state=ADDED, bytes=0, packets=0, duration=31, live
Type=UNKNOWN, priority=100, tableId=c_ingress.get_acl, appId=SiNET, se
lector=[hdr.aid.aid_s=0x20010001000000000000000000000000, hdr.aid.aid_d=0
x20010002000000000000000000000000], treatment=DefaultTrafficTreatment[im
ediate=[c_ingress.allow()], deferred=[], transition=None, meter=[], clear
ed=false, StatTrigger=null, metadata=null]

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Fig. 4. Partial flow entries for user 1 and user 2.

We test the average time it takes for a user to connect to a service after enabling the access control system and compare it to the average connection time when the access control system is not enabled.

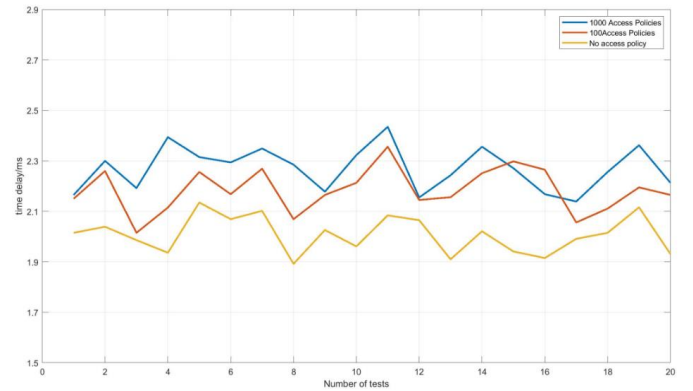


Fig. 5. User communication delay.

Figure 5 compares user communication delay between 1000 access policies and no access policies in the access control system with 1000 access policies. When the access control system is not enabled, the average communication delay for users is 2.01ms. When there are 100 access policies and 1000 access policies in the system, the average communication delay of users is 2.18ms and 2.27ms. The above results show that the access control system has almost no impact on the original communication delay, because in the programmable platform, the user's access policy is sent to the programmable switch in the form of a flow table, which prevents the switch from frequently querying the access policy to the policy matching point when implementing the access policy, and improves the execution efficiency of the system.

In order to ensure that this access control system can still be used usually under high concurrency conditions, our JMeter stress test tool in the environment shown in Figure 1 enables a device to emit a large number of packets to simulate the scenario of large-scale terminal concurrency.

TABLE II. POLICY QUERY DELAY IN ACCESS CONTROL SYSTEM

Number of users	This paper (ms)	Traditional SINET (ms)
100	35.61	103.52
200	35.92	104.11
300	36.45	104.92
400	37.21	105.72
500	37.96	106.54

Table II shows the delay of ASR sending access policy queries in the access control system for 100 access policies. From the table analysis, with the increase of the number of users, the delay of the system policy query has increased to a certain extent, but the increase is tiny. It shows that the number of users has a particular impact on network communication, but the function of the access policy query is stable. In this paper, due to the use of programmable switches to execute policy queries, the query latency is significantly reduced compared with the traditional SINET policy queries.

V. CONCLUSION AND FUTURE WORK

This paper proposed an access control and control system based on the programmable platform through the research of SINET architecture and attribute-based access control mechanism under the programmable platform. By combining the identity and attributes in the identification network, users can conduct security control when they access the network using network devices and achieve fine-grained control for users and services. At the same time, according to the above design, this paper implemented and validated the functions. The results show that this mechanism can realize the refined user management based on access identification and user attributes, prevent unauthorized users from illegally accessing the core network of the identification network, and ensure the security control of the identification network.

In the future, our work can be divided into two aspects. One is to study the policy management system and dynamically generate access policies through algorithms. The other is to complete more complex user behavior analysis based on machine learning and data mining to achieve more accurate network management.

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